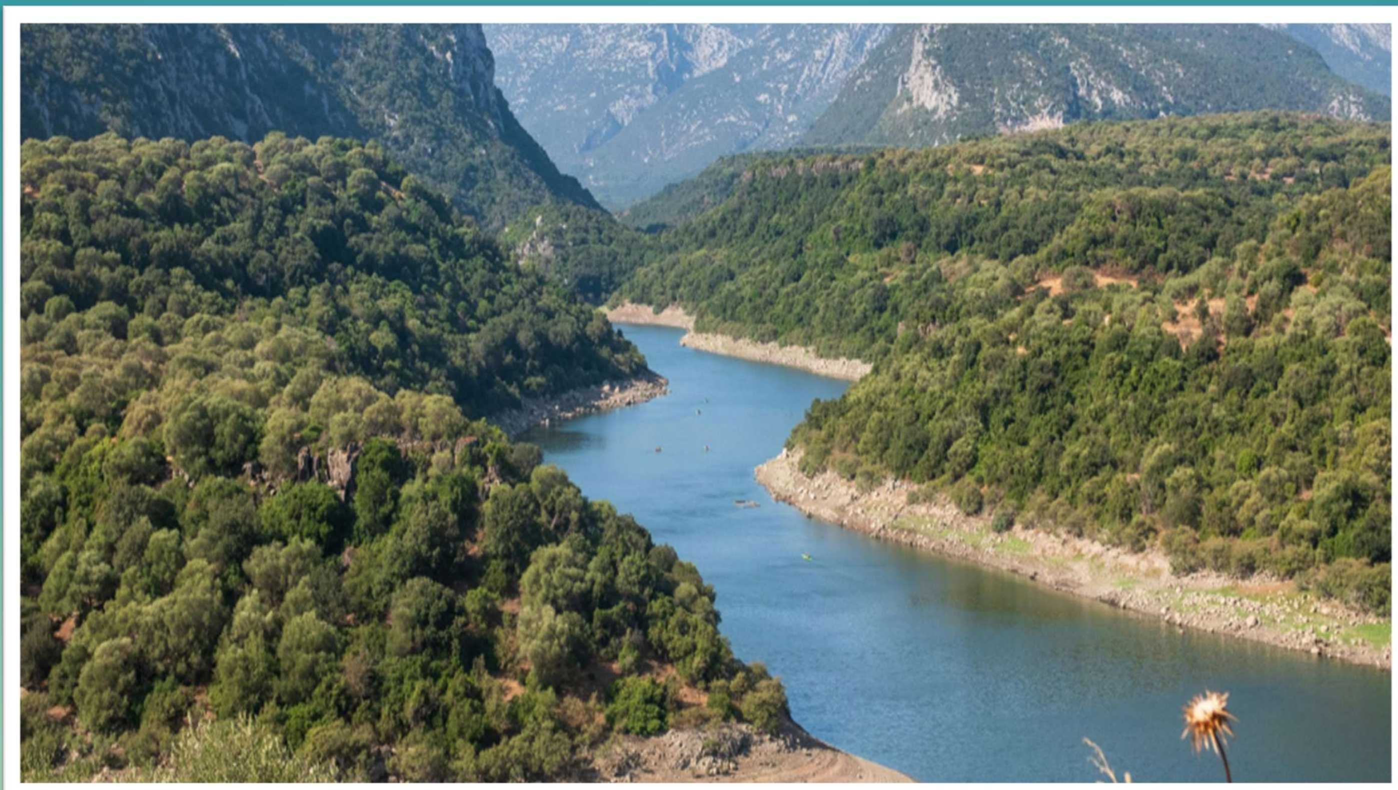

General Field Surveys *(during daylight hours)*

Job Safety Analysis (JSA)

Prepared for Ecology Consulting Pty Ltd

29 January 2026

About this document

This Job Safety Analysis (JSA) has been developed by Aaron Dooley in accordance with Ecology Consulting Pty Ltd Work Health and Safety Management System (WHSMS).

To ensure ongoing compliance and relevance, this JSA will be subject to a bi-annual review by **all staff,** commencing six months from the date of approval.

Document tracking

Version	Unique Identifier	Document Author	Staff consultation completed on
V1.0	EC-JSA-005	Aaron Dooley	29/1/2026

Disclaimer

This JSA has been prepared by Ecology Consulting Pty Ltd to identify and mitigate potential hazards associated with specific ecological fieldwork and operational tasks. The information contained herein is based on current knowledge, site conditions, and standard industry practices at the time of preparation.

While every effort has been made to ensure the accuracy and relevance of this document, Ecology Consulting Pty Ltd makes no warranties, express or implied, regarding the completeness or suitability of the JSA for all circumstances. Site conditions, environmental factors, and operational variables may change, and it is the responsibility of all personnel to remain vigilant and exercise sound judgment in the field.

All personnel must promptly report any new or unforeseen hazards to their designated field lead, followed by notifying the office.

About Ecology Consulting

Accredited

We are NSW-accredited Biodiversity Assessors, licensed to do scientific fieldwork in NSW, and accredited with the national Land for Wildlife program.

Local knowledge

We specialise in the ecosystems of southern NSW, so we can survey, assess, and document your site efficiently and accurately.

Solutions - Not just science

We have helped hundreds of landholders find information, contacts, events, and funding to achieve better environmental outcomes.

Support for major projects

We partner with business, government, and not-for-profit organisations to deliver major construction, species conservation, and site rehabilitation projects.

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1 Introduction

1.1 Purpose

A **Job Safety Analysis (JSA)** is a structured risk management tool used to identify, assess, and control hazards associated with specific tasks, particularly those classified as **high-risk work**. Its primary purpose is to ensure that all personnel understand the risks involved in a job and are equipped with the knowledge and resources to perform it safely.

The key components of a JSA are referenced below.

1.1.1 Hazard Identification

The JSA breaks down a job into individual steps and examines each for potential hazards, physical, environmental, biological, chemical, or procedural. This proactive approach helps prevent incidents before they occur.

1.1.2 Risk Assessment and Control

Each identified hazard is assessed for its likelihood and severity. Appropriate control measures—such as engineering solutions, administrative actions, or personal protective equipment (PPE)—are then implemented to reduce the risk to an acceptable level.

1.1.3 Safe Work Planning

The JSA provides a clear and structured plan for performing the job safely. It ensures that all team members understand the correct procedures, safety requirements, and emergency protocols before work begins.

1.1.4 High-Risk Work Compliance

JSAs are mandatory for the following Ecology Consulting Pty Ltd high-risk activities, such as:

- Working at heights
- Remote or isolated fieldwork

- Handling hazardous substances
- Fauna Handling
- Confined space entry
- Working in bodies of water
- Working on clearing supervision projects
- Other high-risk activities

1.1.5 Communication

JSA's serve as a communication tool between all staff, and subcontractors. Reviewing the JSA as a team ensures everyone is aware of their safety expectations.

1.1.6 Revision of JSA's

JSA's must be reviewed by all Ecology Consulting staff **biannually** and whenever significant changes occur in work practices, legislation, or site conditions. Any updates will follow the formal consultation process to ensure staff input and compliance. Updated versions must be communicated to all team members before implementation.

1.1.7 Incident Prevention

By addressing hazards before work begins, JSAs significantly reduce the likelihood of workplace injuries, equipment damage, and environmental harm. They also contribute to a culture of safety and continuous improvement.

1.1.8 Regulatory Compliance

JSAs help meet legal obligations under WHS legislation and demonstrate due diligence in managing workplace risks.

1.1.9 Report of Incidents, Accidents or Near Misses

All incidents, accidents, or near misses must be reported to the **WHS Officer (Tony)** within **immediately** via **phone or email**. This ensures timely investigation, corrective actions, and compliance with company safety protocols.

Note: Notifiable incidents must be reported to SafeWork NSW or WorkSafe ACT immediately by phone, followed by written notice within 48 hours.

1.2 How to use

This document **must** be read **before commencing** fieldwork and used during field activities to support safe decision-making.
It is designed to be used alongside Ecology Consulting's WHS Management System and project-specific WHS documentation.

1.3 Relationship Between JSA, SWMS and ERP

1.3.1 Job Safety Analysis - *(JSA)*

A JSA breaks a task into steps, identifies hazards, assesses risk, and applies controls using the Hierarchy of Controls.
It covers task-level hazards such as vehicle checks, working near water, environmental exposure, remote work, and general survey risks.
It must be reviewed by the field team before work starts and updated dynamically if new hazards emerge

1.3.2 Safe Work Method Statement - *(SWMS)*

A SWMS is **mandatory** for **high-risk construction** work under WHS legislation *(for example., working near traffic, in or over water, confined spaces, asbestos disturbance, working at heights)*.

A SWMS documents step-by-step methods, required competencies, PPE, exclusion zones, rescue procedures, and verification controls.
If a task in this JSA meets the definition of high-risk work, the relevant SWMS must also be followed.
A list of the mandatory high-risk construction activities can be found on the SafeWork NSW website (*https://www.safework.nsw.gov.au/your-industry/construction/construction*).

1.3.3 Site-Specific Emergency Response Plan- *(ERP)*

The ERP outlines the emergency procedures, communication protocols, key emergency contact information, medical facilities details for each project.
It must be prepared prior to mobilisation, reviewed by all field staff, and kept accessible throughout field operations.
The JSA complements, but **does not** replace the ERP

2 Pre-Mobilisation Controls *(applies to all field activities/tasks)*

These controls must be completed before any travel, field deployment, or on-site commencement. They replace repeated pre-travel content previously embedded in individual task sections. All tasks in this JSA now cross-reference this subsection.

2.1 Pre-Field Meeting Requirements

All staff attending the project must:

- Review the scope of works, field plan, survey methodology and attend a pre-field meeting with the Project Lead/Field Lead and Project Manager.
- Confirm access requirements, keys/codes, induction status, and property boundary constraints.
- Discuss anticipated WHS hazards including weather, terrain, fatigue, communications, and isolation risks.
- Identify staff medical considerations privately with the Field Lead.
- Ensure all attending staff are competent, medically fit, and aware of their responsibilities.

2.2 Pre-Travel Vehicle Inspection

- Complete the Pre-Travel Vehicle Safety Checklist *(Teams - EC_WHS_FORM).*
- Verify presence and condition of:
    - First aid kit, fire extinguisher, spare tyre, tyre kit, high-vis vest, torch, water, recovery gear *(if applicable).*
- Conduct checks on flat ground, apply handbrake, and chock wheels if necessary.
- Perform only visual electrical inspections; do not handle exposed wiring. Tag-out faulty vehicles.
- Use correct manual handling practices when lifting bonnets, gear, or spare tyres.
- Inspect for fuel, coolant, or chemical leaks; do not drive until repaired

2.3 Pre-Travel Environmental & Hazard Checks

Immediately before departure, all teams must check:

- The Bureau of Meteorology (BOM) Weather Website / App *(warnings, rain, storms, temperature*).
- The Hazards Near Me App *(hazard activity, alerts).*
- The Rural Fire Service (RFS) Website / App *(fire behaviour, warnings and fire rating).*
- The Live Traffic NSW Website / App *(road closures, incidents, flood impacts).*
- The Emergency+ App *(has been installed on your device).*
- The Lightning App *(storm proximity).*
- The Air Quality NSW *(AQ)* Website *(smoke/airborne particulates).*
- The State Emergency Service Website *(for information of flooded area).*
- The Essential Energy Website *(Live storm tracker).*
- Your telecommunications network service providers service coverage map.
    - Telstra
    - Optus
    - Vodafone
    - TGP

Do not travel into areas with:

- Emergency Warning or Watch & Act fire alerts
- Hazardous or Very Poor air quality
- Active storm warnings or lightning
- Flooded roads

If conditions change, stop, reassess, and notify the Communication Officer if it is unsafe to~~o~~ proceed.

2.4 Communication & ERP Readiness

Before mobilisation:

- Review the Site-Specific ERP
- Confirm emergency contacts, evacuation routes, nearest medical locations, and meeting points.
- Test communication devices *(for example, mobile, UHF, In Reach).*
- Agree on check-in intervals

3 Risk Assessment Guidance

When assessing raw and residual risk, please refer to the **Appendix 1, "5 x 5 risk matrix"** provided below.

When implementing control measures, it is essential to apply the **Hierarchy of Controls.** This hierarchy should always be approached from the top down, prioritising the most effective controls first:

- **Elimination** - Remove the hazard entirely.
- **Substitution** - Replace the hazard with a safer alternative.
- **Engineering Controls** - Isolate people from the hazard.
- **Administrative Controls** - Change the way people work.
- **Personal Protective Equipment** - Protect the worker with equipment.

Starting from the top ensures that risk is managed as effectively as possible, reducing reliance on less effective measures such as PPE.

4 JSA

4.1 JSA template

Activity/Task	**General Ecological Field Surveys** *(during the day only)*
Location	Ecology Consulting Project Sites

Risk Assessment				
Task - Steps	**Hazard /Risk**	**Raw Risk rating**	**Control**	**Residual Risk**
Schedule and attend pre-field meeting	**Hazards** • Lack of understanding of the scope of works, project-specific access, attending personnel, and WHS hazards. **Risks** • Injury to staff *(physical and psychological).* • Personal injury or vehicle damage • Unknown WHS and site-specific information.	**High (12)**	**Addressing the risk of project scopes of work.** • Ensure all staff attending the pre field meeting and review documents prior to the field work. **Address the lack of project access information / unclear site entry requirements.** • Project Lead to confirm site access arrangements with the client or landholder before the pre field meeting. • Check that all staff have correct access permissions, keys, codes, induction certificates. • Review all property boundaries, restricted zones, and no-go areas with the team.	**Medium (6)**

			• Discuss any seasonal constraints, traffic arrangements, or special access considerations. • Ensure all vehicles attending the field site are fit for purpose and suitable for the terrain. • Input and the opportunity for feedback from all team members is essential in addressing issues. **Addressing other WHS risks.** • Discuss mental load and fatigue management (*e.g. travel time, weather, phone service, isolation*). • Confirm all staff attending are competent, briefed, and medically fit for fieldwork.	
Undertake pre travel vehicle safety checklist	**Hazards** • Vehicle failure during travel, leading to stranded workers or roadside hazards. • Vehicle may move unintentionally while inspecting. • Exposure to faulty batteries, electrical components or accessories. • Strains, sprains or injury from lifting items *(for example, bonnets, spare tyres, equipment.)* • Leak from fuel, coolant, oil or stored chemicals. **Risks** • Vehicle breakdown. • Risk of Injury to all involved staff and witnessing by-standers *(including physically and mentally)* • Increase of financial stress on the individual due to repairs.	**Very High (16)**	**Addressing the risk of vehicle breakdown.** • Conduct a pre-mobilisation check using the template in your Microsoft teams form channel. • Ensure vehicle has a: ○ First aid kit ○ Fire Extinguisher ○ High-visibility vest ○ Torch ○ Water ○ Tyre replacement kit and a good condition spare tyre • Flag and report any issues. **Addressing the risk of vehicle roll-back.** • Conduct checklist on flat, stable ground wherever possible. • Apply handbrake fully before exiting the vehicle. • Engage Park gear or use wheel chocks if on any incline. • Never stand or position yourself in front or behind wheels during checks. • Maintain clear communication with others working around the vehicle. • Ensure the area is free of pedestrians before conducting checks.	**Medium (6)**

			Addressing the risk of electric shock. • Perform visual inspection only, **do not** touch exposed wiring or damaged components. • Ensure all electrical work is done by a qualified mechanic or auto-electrician. • Turn off the ignition and ensure keys are removed before checking under the bonnet. • Report and tag out vehicles with electrical faults–do not operate until repaired. • Always adhere to the manufacturing hand book. **Addressing the risk of manual handling injuries.** • Use proper lifting techniques *(for example, bend knees, straight back, keep load close).* • Ask for help when lifting heavy items such as spare tyres or recovery gear. • Use mechanical aids where *possible (for example, vehicle jacks, lifting ramps)*. • Avoid awkward postures–reposition yourself instead of overreaching. • Store equipment in the vehicle to minimise twisting and excessive bending. **Addressing the risk of a chemical spill during inspection** • Inspect the ground for visible leaks before starting the vehicle. • If reasonably practical carry a spill kit appropriate to the vehicle and materials on board. • Clean spills immediately using proper PPE and disposal procedures as per the chemicals Safety Data Sheet (SDS). • **Do not** drive vehicle until inspected and the issues has been repaired.	
Undertake pre travel site/enroute safety/ hazard checks	**Hazards** • Lack of WHS planning. • Bushfires / Smoke. • Electrical Storms / Lightning.	**Very High (16)**	**Addressing the risk of decrease emergency response times and risk of physical or psychological injures.** • Complete the Site-Specific ERP prior to field mobilisation,	**Medium (6)**

	• High Rainfall / Floods. • Other Environmental Hazards *(for example, heat, wind, dust, fallen trees, poor visibility, etc.).* **Risks** • Decrease emergency response times . • Risk of physical or psychological injury to all staff. • Increased likelihood of vehicle incidents. • Exposure to unknown or changing WHS hazards. • Delays, entrapment, or isolation due to severe weather or environmental conditions.		Ensure all project staff review and understand the ERP before travelling. • Confirm emergency contacts, evacuation routes, nearest hospitals, and communication procedures. • Complete the pre-mobilisation checks. **Addressing the risk of injury due to bush fire / smoke** • Refer to Section 2 – "Pre-mobilisation control checks" for, all the following ○ Electrical storm, ○ Lightning, ○ High rainfall, and flooding, and ○ Other environmental hazard requirements. • These must be followed before travelling and continuously monitored while on site. • **Do not** travel into areas under Emergency Warning, Watch and Act, or significant smoke hazard. • **Do not** travel into areas with a "*Very Poor or Hazardous"* air quality rating. • Always monitor changes during travel and stop if visibility deteriorates. • Carry P2/N95 masks if conditions involve smoke exposure. **Addressing the risk of injury due to electrical storms, and** **addressing the risk of injury due to high rainfall / flooding** • Refer to Section 2 – "*Pre-mobilisation control checks*" for, all the following ○ Electrical storm, ○ Lightning, ○ High rainfall, and flooding, and ○ Other environmental hazard requirements. These must be followed before travelling and continuously monitored while on site. **Addressing the risk of injury due to other WHS hazards**. • Monitor for heatwaves, high winds, low visibility, fallen branches, or road obstructions.	

			• Adjust travel times or postpone fieldwork if environmental warnings are issued. • Ensure vehicles carry additional water, first aid, and charged communication devices. **Notify the Communication Officer, if it is unsafe to proceed**	
Driving / Travelling	**Hazards** • Bushfires / Smoke • Electrical Storms / Lightning • High Rainfall / Floods • Other Environmental Hazards (heat, wind, dust, fallen trees, poor visibility, etc.) • Fatigue • Wildlife • Road conditions • Other road users **Risks** • Decrease emergency response times • Risk of physical or psychological injury to all staff • Increased likelihood of vehicle incidents • Exposure to unknown or changing WHS hazards • Delays, entrapment, or isolation due to severe weather or environmental conditions	**Very High (16)**	**Addressing the risk of injury due to environmental hazards** • Refer to Section 2 – "Pre-Mobilisation check controls" for environmental hazard management requirements • Refer to Appendix 2 – "Bushfire Safety" **Addressing Fatigue.** • Review Journey Management Plan (JMP) prior to travelling (if applicable) (Click here for a template example) • Conduct self-assessment for fatigue before travel. • Adhere to rest breaks every 2 hours; avoid long days followed by long drives. • Ensure you have a restful night's sleep prior to driving • Ensure you have had something to eat, and your hydration level is good. (Click here for Hydration level information) • Do not travel if tired, unwell, or after extended work hours. • Use a second driver for long or remote trips. • Notify the Communication Officer that is it is unsafe to proceed OR you are requiring additional accommodation. **Addressing the risk of driving related injury due to wildlife.** • Reduce speed in dawn/dusk periods and in wildlife-dense areas. • Use high beam when safe to increase detection distance. • Remain alert for animals near road edges. • Stop work or delay travel if wildlife activity is high *(for example, after rain).* **Addressing the risk of driving related injury due to the road conditions and other road users.**	**Medium (6)**

			• Review road conditions via Live Traffic NSW and local council updates. • If reasonably practical use 4WD where required and confirm roads are suitable for the vehicle type. • Avoid damaged, waterlogged, or unmaintained roads. • If reasonably practical carry recovery equipment when remote travel is required. • Maintain safe stopping distances and drive safely. • Observe and adhere to speed limits and adjust for weather or traffic changes. • Be aware of heavy vehicles, roadworks, school zones, and agricultural machinery. **Notify the Communication Officer, if it is unsafe to proceed**	

Undertaking general ecological field work in a bushfire-prone location during bush fire season	**Hazards**
	• Bushfire activity
	• Smoke / Poor Air Quality
	• Electrical Storms / Lightning
	• Poor WHS planning
	• Increase in fire fuel load
	• Rapidly changing weather conditions
	• Sudden road closures
	
	Risks
	• Severe burns or fatality from fire front exposure
	• Entrapment due to fast-moving fire or blocked escape routes
	• Damage to vehicles, equipment, or infrastructure
	• Respiratory irritation or breathing difficulty
		Extreme
		(20)
			Addressing the risk of injury while undertaking surveys in a bushfire-prone location during mid-late Spring and Summer.
			• Refer to Appendix 2 – "*Bushfire Safety* ", for bushfire-related controls, including,
			  ○ Pre-travel requirements,
			  ○ On-site behaviour,
			  ○ cease-work triggers,
			  ○ refuge zones, and,
			  ○ Evacuation procedures.
			Note: These controls apply to all project sites during bushfire season or when fire danger ratings are elevated.
				Medium
				(8)

	• Increased smoke production resulting in poorer visibility • Eye irritation and fatigue impacting concentration • Increased likelihood of a fire starting if there is a lightning strike • Falling branches/debris during storm conditions • Loss of communications or navigation due to storm impacts • Delayed or ineffective emergency evacuation • Staff exposed to unplanned hazards due to lack of information • Increased stress or fatigue leading to poor decision-making • Faster and more intense fire spread increasing burn risk • Sudden heat or wind changes increasing fire behaviour risk • Higher likelihood of slips, trips, falls in unstable conditions			

Working in remote areas/ areas of poor communication service	**Hazards**
	• Lack of mobile reception / limited communication coverage
	• Remote location / isolation from medical help
	• Difficult terrain and poor access routes
	• Environmental hazards (heat, storms, wind, lightning, bushfire)
	• Failure of communication devices
	• Working alone or in small teams
		Very High (16)	**Addressing the risk of injury when working in a remote / isolated project location.**
			• Refer to the relevant "EC JSA Remote and Isolated Work v1"
			  ○ Always work in pairs, -working alone is not permitted in remote or low-reception environments.
			  ○ Complete and review the Site-Specific ERP before travel, including emergency contacts, evacuation routes, and rendezvous points.
			  ○ Establish mandatory check-in intervals with the Communication Officer and follow the escalation procedure if a check-in is missed.
			  ○ Carry at least two forms of communication: mobile + two way + In Reach for no-signal areas. Ensure all are tested before departure.
				High (10)

	• Incorrect or incomplete WHS planning before remote travel **Risks** • Delayed or failed emergency response during injury or incident • Inability to contact support, landholders, or emergency services • Isolation increasing psychological stress and poor decision-making • Extended time without medical treatment if injury occurs (snake bite, fall, heat stress, etc.) • Entrapment or inability to self-evacuate due to distance or terrain • Complications from medical conditions without timely assistance • Becoming stranded or stuck (vehicle bogging, blocked roads) • Delayed rescue if access routes deteriorate due to weather or fire • Increased risk of fatigue or dehydration due to long travel periods • Heat stress, dehydration, hypothermia or exposure illness • Field staff unable to receive warnings of changing weather due to poor signal • Inability to evacuate safely during a storm, bushfire, or rapid weather change • Loss of situational awareness and inability to check-in at required intervals • Inability to coordinate search and rescue if staff go missing or injured		○ Confirm staff are competent, medically fit, and aware of isolation risks before mobilisation. ○ Carry essential equipment: adequate water, food, remote first-aid kit, maps/GPS, and spare clothing. ○ Identify difficult terrain, poor access routes, and potential entrapment points before entering the work area. ○ Cease work immediately and notify the Communication Officer if comms fail, weather deteriorates, route access becomes unsafe, or an emergency arises. ○ Ensure team members can recognise early signs of heat stress, hypothermia, fatigue, or medical deterioration and activate ERP procedures if needed. ○ Do not proceed into remote terrain unless all staff have required gear, check-ins are established, and communication equipment is fully operational	

	• No immediate assistance available during medical emergency or threat (fauna, aggression, fall, weather) • Increased psychological stress or panic in an emergency • Higher likelihood of delayed discovery if a staff member becomes unresponsive or lost • Team unaware of emergency response steps, meeting points or check-in requirements • Failure to recognise hazards due to lack of maps, terrain info, or access notes • Staff working in hazardous remote areas without required equipment (In Reach, PLB, water, remote first aid kit)			

Work on, in or adjacent to a road, railway, shipping lane or other traffic corridor that is in use by traffic other than pedestrians	**Hazards**
	• Live traffic operating near the work area
	• Mobile plant and machinery movement in proximity to workers
	• Unsafe or restricted site access (soft edges, drop-offs, poor parking areas)
	• Limited line-of-sight when entering or exiting the road/rail corridor
	• Unauthorised entry of vehicles or pedestrians into the work zone
	• Unexpected train movement in rail corridors
	• Lack of required rail/traffic safety inductions
	• Environmental conditions (fog, rain, dust, wind, storm activity) reducing visibility
	• Miscommunication with traffic controllers, rail officers, or work team
		Very High
		(16)
			Addressing the risk of injury due to working on, in or adjacent to a road, railway, shipping lane or other traffic corridor that is in use by traffic other than pedestrians.
			• Refer to the relevant Safe Work Method Statements 15 (SWMS) for "*Work on, in or adjacent to a road, railway, shipping lane or other traffic corridor that is in use by traffic other than pedestrians* ".
			Note: This is a high-risk activity under the Work Health and Safety Regulation 2017 (NSW), therefore a SWMS is required.
			• Ensure all staff have completed the Daily Toolbox Talk prior to commencing any work in or near traffic corridors.
			• All staff must have the following accreditation.
			  ○ CPCWHS1001 – Prepare to Work Safely in the Construction Industry.
			• In addition, if you are working on, in a railway corridor, you must have completed the following training.
			  ○ TLIF0020 - Safely access the rail corridor
			• Field Lead must confirm site-specific hazards, safe parking areas, exclusion zones, entry/exit points, and communication protocols
				High
				(8)

	• Communication device failure (UHF, mobile, two-way)
	Risks
	• Workers struck by passing vehicles due to proximity to live traffic
	• Collision with mobile plant or machinery operating in or near the corridor
	• Vehicle roll-over or bogging when parked on unstable edges or embankments
	• Accidents caused by poor visibility (fog, rain, low light, dust, smoke)
	• Workers or vehicles entering unsafe traffic lanes due to unclear exclusion zones
	• Serious injury or fatality due to unexpected train movement in rail corridor work
	• Traffic incidents caused by insufficient signalling, communication or traffic control
	• Communication failure leading to incorrect instructions, unsafe vehicle movements or delayed emergency response
	• Increased risk of slips, trips, falls, or vehicle accidents due to adverse weather conditions
			with traffic control personnel *(or rail safety officers for rail corridors).*
			• Engage qualified traffic controllers where required and ensure Traffic Guidance Schemes *(TGS)* or rail protection measures are implemented before beginning work.
			• All workers must wear appropriate high-visibility PPE, as required for road or rail corridor work.
			• Ensure communication devices *(two-way radio, mobile, In Reach)* are operational and tested before starting work. [
			• Only enter designated work areas once traffic control measures are verified by the Field Lead during the pre-start drive-through inspection.
			• Maintain clear exclusion zones around live traffic lanes or rail lines; ensure all personnel understand the Danger Zones and safe retreat areas.
			• An active *"Spotter* "must be assigned to monitor for traffic and other hazards while the driver exits the vehicle and during the unloading of equipment.
			• The Spotter must remain within 5-10 metres of the person performing the task at all times.
			• The Spotter must **not** engage in any other activities, including assisting with unloading or handling equipment.
			• The Spotters primary responsibilities include:
			  ○ Monitoring for approaching traffic or hazards.
			  ○ Judging and communicating when it is safe to exit the vehicle and unload equipment.
			• The Spotter must **not** use a mobile phone or any other device while actively spotting.
			• Where applicable *(rail environments),* ensure all staff have completed the required rail safety inductions *(e.g., ARTC, RIW, Martinus).*
			• Stop work immediately if traffic control, visibility, or weather conditions deteriorate to unsafe levels *(fog, heavy rain, high winds, smoke, or lightning).*
			Notify the Communication Officer, if it is unsafe to proceed	

Undertaking general ecological site surveys *(only during the day)*	**Hazards** • Unknown underlying medical issues associated with assisting staff • Extreme weather (heat and cold) • Lack of understanding the scope of works and methodology • Potential vehicle interaction with passing traffic at roadside areas • Unstable ground (especially after rain) • Slips, trips and falls • Flicking of sticks or debris while walking • Previously burnt sites (unstable trees, ash pits, weakened ground) • Loading and unloading equipment • Climbing fencing (only if required) • Silica *(if present onsite).* • Asbestos *(if present onsite).* • Construction site hazards *(if entering or adjacent).* • Confined spaces *(if entry is required)* • Psychosocial hazards *(for example, stress, fatigue, conflict, isolation).* • Presence of known or unknown individuals or aggressive behaviour. • Presence of aggressive or unpredictable animals *(for example, wildlife or livestock).* • Hearing gunshots, explosives or threatening distant activity. • Suspicious or slow-moving nearby vehicles. • Environmental hazards (*for example, wind, storms, lightning, heat, cold, smoke).*	**Very High (16)**	**Addressing the risk of decrease emergency response times and risk of physical or psychological injures.** • Complete all pre-mobilisation checks • Review and monitor all the previously mentioned controls, adjust if needed. **Addressing the risk of injury due to bush fire / smoke** • Refer to Section 2 - "Pre-Mobilisation check controls" for environmental hazard management requirements • Refer to Appendix 2 - "Bushfire Safety" **Addressing the risk of injury due to electrical storms.** • Refer to Section 2 - "Pre-Mobilisation check controls" for environmental hazard management requirements **Addressing the risk of injury due to high rainfall / flooding.** • Refer to Section 2 - "Pre-Mobilisation check controls" for environmental hazard management requirements **Addressing the risk of reduced medical response times and incorrect treatment.** • Medical conditions must be discussed privately between the staff member and the Field Lead before commencing fieldwork. • Health information must be handled in accordance with the Privacy Act and Ecology Consulting's Privacy and WHS Management Policies. • Personal medical details **must only** be shared on a strict need-to-know basis with the Field Lead to support safe planning and emergency response. • All "at-risk" team members must carry required medications (for example, EpiPen, inhaler, etc), and the "at-risk" team member must inform the field lead of their response plan, triggers, treatment procedure and the location of medications. **Examples of "at-risk" health or medical conditions.** ○ **Asthma or respiratory conditions** *(e.g., sensitive to smoke, dust, pollen. Requires Ventolin).* ○ **Severe allergies or anaphylaxis** *(for example, bee/wasp stings, certain foods. Requires EpiPen).* ○ ***Diabetes*** *(risk of low blood sugar. Requires regular breaks/food/insulin management).*	**High (8)**

	• Poor air quality *(for example, smoke, dust, pollen).* • Remote location / isolation from assistance. • Inadequate communication or communication device failure. **Risks** • Unknown. • Worker struck by passing vehicles or machinery. • Serious injury or fatality due to poor visibility around traffic • Slips, trips, or falls causing sprains, fractures or soft-tissue injuries. • Eye injuries or cuts from flying debris. • Impact injury from falling branches or fire-damaged trees. • Ground collapse or stepping into hidden holes in burnt areas. • Back injuries, muscle strain or crush injury from manual handling. • Falls or entanglement when climbing fencing. • Respiratory illness from inhalation of silica dust. • Long-term health risks from exposure to asbestos fibres. • Injury from construction plant, machinery, uneven ground, or falling objects. • Asphyxiation, toxic exposure, entrapment or falls in confined spaces. • Stress, fatigue, reduced alertness or mental overload.		○ **Heart conditions or high blood pressure**. ○ **Epilepsy or seizure disorders**. ○ **Recent injuries** *(for example, sprains, fractures, reduced mobility).* ○ **Mental health or psychosocial** vulnerabilities *(for example, anxiety, PTSD triggers, high-stress sensitivity).* ○ **Heat or cold sensitivity** *(for example, history of heat stress, dehydration, hypothermia).* ○ **Medication-dependent conditions where emergency access to medication is required.** • Field Lead to monitor staff for early symptoms during the shift. **Addressing the risk of Hyperthermia** *(heat).* • Refer also to Section 2 – "*Pre-Mobilisation checks*", for the following, ○ Weather, ○ Heat, and, ○ Other environmental hazard requirements. • Wear lightweight, breathable, long-sleeve clothing, wide-brim hats, and UV-rated PPE to reduce heat absorption. • Apply sunscreen regularly to prevent sunburn. • Increase frequency of hydration breaks and monitor your hydration levels throughout the day. • It is recommendation to add electrolyte solutions or hydration tablets to your fluid in-take to replace salts lost through sweat. • Undertake the physically demanding tasks during cooler morning hours and low-intensity tasks during peak heat periods. • Limit time spent in direct sun. • If reasonably practical rotate field team members regularly during high-temperature periods to reduce heat load and fatigue. • Ensure all staff have access to minimum 10 litres of water for the day and refill points if remote. • Ensure vehicles have functioning air-conditioning	

	• Physical or psychological harm from aggressive people or animals.
	• Threatening behaviour escalating to violence or panic.
	• Bee/wasp stings, spider bites, snake bites, or other zoonotic hazards.
	• Panic or disorientation from gunshots or explosive sounds.
	• Vehicle intimidation *(for example, tailgating, slow passing, stalking behaviour).*
	• Heat stress, dehydration, hypothermia or exposure illness.
	• Lightning strike, falling branches, storm-related injuries.
	• Respiratory issues and reduced visibility from smoke or dust.
	• Delayed emergency response due to remote location.
	• Inability to call for help due to communication failure.
			• Cease work immediately if any worker shows signs of heat exhaustion or heat stroke, activate emergency response procedures.
			• Ensure you have packed adequate clothing to address the project site weather conditions.
			• Ensure you have adequate food and a minimal of 10 litres of water available.
			• Avoid prolonged fieldwork during extreme heat or cold.
			• Adjust work hours to morning/late afternoon.
			• Provide shaded rest areas, warm layers, electrolytes, and hydration break.
			Addressing the risk of Hypothermia *(cold).*
			• Refer also to Section 2 – "Pre-Mobilisation checks", for the following,
			  ○ Weather,
			  ○ Heat, and,
			  ○ Other environmental hazard requirements
			• Ensure all staff wear appropriate PPE, including thermal layers, wind-resistant outerwear, beanies, gloves, and waterproof boots when required.
			• Keep spare dry clothing in vehicles to change into if garments become wet from rain, sweat, or water crossings.
			• Ensure you have adequate food and a minimal of 10 litres of water available.
			• If reasonably practical regularly rotate staff through warming breaks in vehicles during cold or wet conditions.
			• Minimise exposure to high-wind areas *(e.g., ridge lines, open paddocks)* where wind chill significantly increases hypothermia risk.
			• Ensure vehicles are fuelled and able to run heaters if emergency warming is required.
			• Ensure the communication check-in schedule is followed *(using phone, or In Reach device).*	

			• Provide training for staff on recognising and responding to cold-stress symptoms, including practical first-aid steps for mild to moderate hypothermia. • Avoid working alone in remote areas during winter or severe weather conditions. • Postpone or cease work if conditions deteriorate *(for example, sudden cold front, high winds, heavy rain, sleet)*. • Keep staff mobile during fieldwork where possible, minimise static tasks that increase heat loss. • Ensure wet-weather gear *(for example, waterproof jackets,),* are worn when rain is forecast or likely. **Addressing the risk of injury due to incorrect survey techniques.** • Review and confirm correct survey methodology during toolbox talk. • Field Lead must ensure all staff are trained/competent in the method for that survey type. • Staff must re-read field plan, method notes, and SOPs before commencing. • **If unsure**, STOP and clarify with Project Lead before continuing. • Field Lead to undertake quality assurance process during fieldwork to ensure accuracy and consistency, as well at the end of the day. • Report errors immediately to both the Field Lead and the Project Lead so data can be corrected before leaving site. **Addressing the risk of injury due to vehicle, plant and machinery movements, and,** **Addressing the risk of injury due to reduced visibility around roadside traffic.** • *Refer to the relevant SWMS 15 for "Work on, in or adjacent to a road, railway, shipping lane or other traffic corridor that is in use by traffic other than pedestrians".* • Refer to the relevant SWMS 16 for "*Work in areas in which there is movement of powered mobile plant".*	

			Note: This is a high-risk activity under the Work Health and Safety Regulation 2017 (NSW), therefore a SWMS is required. • Wear high-vis PPE and stay alert to traffic conditions. • Position vehicle safely with clear sight distance. • Cease work during fog, heavy rain, smoke, or low light. • Use hazard lights or beacons. • Field lead to reassess visibility conditions frequently. **Addressing the risk of injury due to slip, trips and falls.** • Wear lace-up boots with tread. • Pre-assess terrain before stepping. • Avoid unstable or waterlogged areas. • Maintain situational awareness *(“Look Up - Look Around - Look Down”).* • If there is risk of falling more than two metres, refer to the relevant SWMS 01 for “*Work that involves a risk of a person falling more than two metres (note: in some jurisdictions this is 3 metres)”.* **Note:** This is a high-risk activity under the Work Health and Safety Regulation 2017 (NSW), therefore a SWMS is required. **Addressing the risk of injures due to debris and falling branches in burnt/windy areas.** • Complete all pre-commencement checks. • Conduct pre-entry visual assessment. • Assess vegetation density before walking. • Wear safety glasses and hard hats in heavy scrub, windy areas or areas where a previous fire has occurred. • Maintain spacing between team members. • **No** work within 30 m of large burnt trees. • Cease work during winds >25 km/h. • Wear hard hat where tree-fall risk exists. • Avoid smouldering or unstable surfaces. • Reassess site if subsidence detected.	

			Addressing the risk of musculoskeletal injures • Always u~~U~~se correct manual-handling techniques *(for example, bend knees, keep back straight, keep load close to body).* • Apply a two-person lift for items heavier than 20 kg or awkward/bulky equipment. • Ensure all field equipment is secured safely in vehicles to prevent unexpected movement or lifting strain. • Avoid twisting your body while carrying loads; reposition feet instead. • Break equipment loads into smaller trips where possible to avoid overloading. • Use mechanical aids for *(for example trolleys, lifting straps, jacks, or slide-boards)* when~~re~~ reasonably practicable. • Conduct a visual assessment of the lifting environment for slip hazards, uneven ground, or obstructions. • Encourage staff to warm up/stretch before heavy manual tasks to prepare joints and muscles. • Report any pain, early discomfort, or fatigue immediately to the Field Lead. • Provide regular rest breaks during repetitive lifting tasks to prevent fatigue-related strain. **Addressing the risk of injury due aggression, threatening behaviour, or confrontation from stakeholders, landholders, or members of the public:** • Follow the "Six-Step Conflict Response Framework " **Addressing the risk of injury caused by an aggressive or unpredictable animal.** • Keep safe distance; never approach. • Identify escape routes. • Use mobile phones/In Reachs for team communication. **Addressing the risk of injury due to suspicious or intimidating vehicle behaviour, and the risk of injury due to hearing gunshots or explosive noises causing panic.**	

			• Stop work immediately and reassess the situation to determine the location and nature of the hazard. • Move the team to a safe, visible area away from potential threat zones. • If required, move directly to the vehicle, lock all doors, and remain inside until it is safe to exit. • If safe to do so, relocate or leave the site and travel to a known safe location such as a public area or nearby town. • Contact police on triple zero (000), if behaviour appears threatening, persistent, or suggests risk of harm. • Notify the Communication Officer and WHS Officer immediately via mobile or In Reach device. • Maintain communications until the situation is resolved and further instructions are provided. • Do not return to the site until the hazard is cleared, the situation has been investigated, and the team feels safe to proceed **Addressing the risk of injury due to falls or entanglement when climbing fencing.** • Only climb fences if absolutely required. • Create a safe gap and cushion barbs with hats/backpacks. • Avoid rusted or damaged sections. **Note:** Staff are accountable of ensuring their vaccinations are up to date. **Addressing the risk of injury due to working on a construction site.** • All team members must complete all required pre commencement checks prior to attending the construction site *(e.g., Daily Toolbox WHS Form, Pre Travel Checks, communication checks).* • Before commencing any field work, all team members must have successfully completed the following. ○ *CPCWHS1001 - Prepare to Work Safely in the Construction Industry.*	

			• For ecological scopes of work on a construction site within the Australian Capital Territory, all team members must have additionally completed the following accredited courses: ○ *10314NAT - Asbestos Awareness, and,* ○ *0830NAT - Silica Awareness* • Follow all site induction requirements, including site-specific WHS procedures, traffic management instructions, and any rules implemented by the Principal Contractor or Site Supervisor. • Wear all mandatory construction site PPE, including (*but not limited to)*: ○ High visibility clothing ○ Hard hat ○ Steel capped boots ○ Any additional PPE required by the site (e.g., gloves, safety glasses, hearing protection). • Maintain exclusion zones around mobile plant and heavy machinery and never enter live plant operating zones unless authorised and accompanied by site personnel. • Always follow the directions of the Site Supervisor or Principal Contractor and comply with all signage, barriers, and restricted access zones. **Addressing the risk of injury or exposure to asbestos while actively working on a project site** *(observation-only ecological surveys).* • Contact the client or Principal Contractor before scheduling fieldwork to confirm any known site contamination, including asbestos or silica. • Principal Contractor **must inform** Ecology Consulting of any known contaminants and the controls they have implemented. **If suspected asbestos is found:** • Stop work immediately • Do not disturb the material. • Isolate or move away from the area. • Notify the Field Lead and Principal Contractor immediately.	

			• Do not re-enter the area until instructions are provided by the Principal Contractor. • Request and review all relevant documentation such as the Asbestos Register and Hazardous Materials Management Plan. • Ecology Consulting team members **will not** disturb or handle asbestos or silica-containing materials under any circumstances. • Refer to the relevant SWMS 04 for "*Work that involves, or is likely to involve, the disturbance of asbestos (Observation Only)*". **Note:** This is a high-risk activity under the Work Health and Safety Regulation 2017 (NSW), therefore a SWMS is required. **Addressing the risk of injury or exposure to silica while actively working on a project site** *(observation-only ecological surveys).* • Confirm with the Principal Contractor whether any silica-generating activities *(stone cutting, concrete cutting, grinding, drilling)* will be occurring during the visit. • Review site induction materials and the Principal Contractor's Silica Dust Management Plan. • Confirm all staff have successfully completed the following ○ *0830NAT- Silica Awareness training,* before entering an active construction site. • Ecology Consulting staff **must not** enter areas where stone or concrete cutting is occurring. • Maintain a safe exclusion distance as established by the Principal Contractor, typically beyond visible dust drift, and outside any barricaded or flagged silica control zones. • **Never** walk behind, near or through cutting operations, even if dust suppression *(for example, -water suppression)* is being used. • Do not approach, lean on, or touch any surfaces that may be contaminated with silica dust. • Avoid downwind locations where airborne dust may travel. • If cutting unexpectedly begins nearby, stop work immediately and relocate upwind or to a designated safe zone. • Where dust drift is ~~possible~~possible, but work zones cannot be avoided, wear:	

			○ P2 respirator, ○ Safety glasses or goggles, ○ Long sleeves and long pants, and ○ Enclosed boots. • PPE requirements must follow the Principal Contractor's instructions for working near silica-affected areas. • Report unexpected silica exposure, uncontrolled dust, or entry into an unmarked hazard zone to: ○ Field Lead ○ Principal Contractor ○ Communication Officer/Project Manager ○ WHS Officer • If exposure is suspected, complete required WHS reporting procedures and await direction before returning to work. **Addressing the risk of injury if working in a confined space.** • Refer to the relevant SWMS 06 for "Work in or near a confined space". • Refer to the relevant SWMS 08 for *"Work in or Near a Tunnel (Ecological Survey Activities)".* **Note:** This is a high-risk activity under the Work Health and Safety Regulation 2017 (NSW), therefore a SWMS is required. **Notify the Communication Officer, if it is unsafe to proceed**	
Undertaking general ecological surveys in or near a water body	**Hazards** • Slippery Edges • Water • Unknown hazards **Risk** • Drowning • Sprains, fractures, or soft-tissue injury. • Impact injuries from hitting submerged objects or sharp edges.	**Extreme (20)**	**Addressing the risk of drowning.** • Refer to the relevant SWMS 18 for *"Work in or near water or other liquid that involves a risk of drowning"*. **Note**: This is a high-risk activity under the Work Health and Safety Regulation 2017 (NSW), therefore a SWMS is required. • Mandatory PFD/life vest when working in water. • Buddy system: one worker in water, one on bank at all times. • Identify safe entry/exit points before starting work. • Keep throw ropes, throw bags, and rescue devices immediately accessible.	**High (12)**

	• Equipment loss or being pulled off balance while carrying gear. • Inability to self-rescue due to steep banks, slippery edges, entanglement, or fatigue. • Hypothermia • Foot entrapment in debris, tree roots, or underwater voids leading to potential drowning. • Injury from sharp or hazardous submerged objects (*metal, glass, timber).* • Exposure to contaminated water, causing illness or infection. • Unexpected wildlife encounters *(snakes, eels, insects)* leading to bites or stings. • Ground collapse or bank instability, especially after rain or flooding		• Avoid water deeper than knee-height where visibility is poor or footing unknown. • Stop work during fast-flowing or rising water conditions. **Addressing the risk of musculoskeletal injures.** • Review and monitor all the previously mentioned controls, adjust if needed. **Addressing the risk of impact injuries from submerged objects or sharp edges.** • Visually inspect water and use a pole or stick to check for hazards before entry. • Move slowly and deliberately; do not step where visibility is poor. • Wear protective boots and waders to reduce injury from sharp objects. • Avoid areas with visible scrap material, broken timber, or debris accumulation. **Addressing the risk of the inability to self-rescue** *(for example, steep banks, slippery edges, entanglement, fatigue).* • Select low-risk access points with gentle slopes and firm footing. • Avoid thick vegetation, logs, or debris that may cause entanglement. • Use ropes or grab-lines to assist entry/exit where slope is moderate. • Ensure workers rotate frequently to reduce fatigue. • Keep a designated rescuer on standby during water entry tasks. **Addressing the risk of Hypothermia.** • Review and monitor all the previously mentioned controls, adjust if needed. **Addressing the risk of potential health issues caused by exposure to contaminated water.** • Avoid direct contact where contamination risk is high *(for example, stock access, sewage, stagnant water*). • Wear waterproof gloves and avoid touching face during work.	

			• Do not enter water with open cuts or wounds. • Wash hands thoroughly before eating, drinking, or leaving site. • Carry antiseptic wipes and relevant first-aid supplies. **Addressing the risk of injury due to encounters with wildlife.** • Conduct a visual sweep before entering long grass or water. • Avoid placing hands under rocks, logs, or submerged objects. • Wear long pants, gloves, and waders for protection. • Pause work if wildlife is present until it is safe to continue. • All staff must have successfully completed the following training prior to undertaking **any** field work. ○ *HLTAID013 - Provide First Aid in Remote or Isolated Sit, and* ○ *HLTAID009 - Provide Cardiopulmonary Resuscitation* • Carry appropriate first-aid equipment including snakebite bandages. **Notify the Communication Officer, if it is unsafe to proceed**	
Aquatic fauna handling, relocation and euthanasia	**Hazards** • Aquatic ~~F~~fauna • Water bodies **Risks** • Soft-tissue injury, bites • Psychological injury *(associated with euthanasia or stressful handling)* • Physical injury *(claws, bites),* envenomation • Drowning	**Very High (20)**	**Addressing the risk of injury in general.** • Refer to relevant governing documents for all activity-specific requirements, including SWMS, SOPs, and JSAs. • Ensure sufficient trained and competent fauna handlers are assigned to the task. • Prior to field deployment, plan capture and handling steps to minimise stress on fauna and reduce risk to personnel. • Identify the nearest local veterinarian with relevant species expertise in the Site-Specific ERP. **Addressing the risk of soft-tissue injuries.** • Review and monitor all the previously mentioned controls, adjust if needed. **Addressing the risk of psychological injury** *(for example, handling stress and injury response).* • Staff must be trained and briefed on handling methods *(where applicable).* • Conduct toolbox briefings before work to discuss emotional impacts, difficult scenarios, and personal readiness.	**High (10)**

			• Provide clear instructions on humane procedures to minimise distress to fauna and staff. • Conduct end-of-day debriefing. **Addressing the risk of physical injury** *(claws, bites), envenomation.* • Always prioritise human safety over data collection or species identification • Field staff must be trained in envenomation first aid, including recognition and treatment of allergic or anaphylactic reactions. • Only personnel with appropriate fauna-handling training and qualifications may handle fauna. • Handling must be avoided unless necessary for relocation or welfare purposes; observational identification preferred. • Under no circumstances may venomous fauna — including venomous snakes or platypus, be handled without specific, verified training. • Use tools (nets, buckets, containers) to avoid direct contact where possible. • Wear appropriate PPE: gloves, gaiters, arm protection where relevant. • Staff must maintain situational awareness and cease handling immediately if safety cannot be guaranteed. • A trained spotter must be present to assist with safe handling. • First-aid kits must be fully stocked and contain supplies for bites, scratches, and envenomation. • Maintain hygiene protocols: disinfect equipment and PPE between individuals where contamination risks exist. • Contact the WHS Officer for additional information about support services. **Addressing the risk of drowning** • Review and monitor all the previously mentioned controls, adjust if needed **Stop work immediately during fast-flowing or rising water conditions and notify the Communication Officer, if it is unsafe to proceed**	

4.2 Other Matters for Consideration

Dynamic Hazard Management:

- If new hazards are identified at any stage of the works, the JSA must be immediately updated and the risk assessment amended to reflect the newly identified hazards. All team members must review and acknowledge the updated JSA before continuing work.

Team Awareness and Readiness:

- All personnel must review the JSA, relevant SWMS, and activity-specific procedures before commencing field work.

Weather and Environmental Monitoring:

- Weather conditions *(including wind, lightning, extreme heat, and extreme cold)* must be continuously monitored. Work must cease if conditions become unsafe or visibility deteriorates to a level that presents risk.

Cease Work Protocol:

- If conditions become unsafe or workers have concerns about proceeding, the Communication Officer must be notified immediately and work suspended until hazards are resolved or conditions improve.

Emergency Preparedness:

- A fully stocked first aid kit, snake-bite kit, and emergency contact list must be readily accessible at all times.

Fatigue and Hydration Management:

- Appropriate fatigue-management practices must be implemented. Workers must take regular rest breaks, maintain adequate hydration, and self-monitor for signs of fatigue.

Communication Requirements:

- All communication devices must be tested prior to commencing work. Backup communication systems should be available in remote or isolated locations.

Animal Welfare & Ethical Handling:

- Despite thorough precautions, occasional minor injuries to fauna may occur during necessary capture or restraint. Only trained and competent fauna handlers may undertake these tasks. Ecologists will assess basic health indicators such as trauma and responsiveness.

- If an animal sustains a minor injury (e.g., superficial abrasion) but remains alert and active, and immediate release is in its best welfare interest, it will be returned to suitable natural cover at the capture site.

- Animals showing signs of injury, illness, shock, or compromised mobility will be transported to a licensed veterinarian in accordance with the Site-Specific ERP.

Incident and Near-Miss Reporting:

- All incidents, accidents, near misses, and environmental concerns must be reported immediately to the **WHS Officer** and documented according to company procedures.

Clarification of Ecological Constraints:

- Any uncertainty regarding ecological constraints, fauna handling requirements, or species-specific protocols must be clarified with the **Project Lead Ecologist** before work proceeds.

Appendix 1: 5 x 5 Risk Matrix

5 x 5 Risk Matrix						
		←←	Consequence *(severity)*		→→	
		1. Insignificant	2. Minor	3. Significant	4. Major	5. Severe
↑↑ Likelihood	5. Almost Certain	Medium 5	High 10	Very High 15	Extreme 20	Extreme 25
	4. Likely	Medium 4	Medium 8	High 12	Very High 16	Extreme 20
	3. Moderate	Low 3	Medium 6	Medium 9	High 12	Very High 15
	2. Unlikely	Very Low 2	Low 4	Medium 6	Medium 8	High 10
↓↓	1. Rare	Very Low 1	Very Low 2	Low 3	Medium 4	Medium 5

How to use the 5 x 5 Risk Matrix	
Multiplying likelihood by severity (e.g., 2x4=8) gives a score, leading to risk categories like:	
Low *(for example, 1-8)* :	Acceptable, maintain controls
Medium *(for example, 9-15):*	Tolerable, review for improvements
High *(for example, 16-25):*	Unacceptable, immediate action required.

Risk Matrix Definitions	
Likelihood	*The likelihood (x-axis) pertains to the extent of how likely it is for the risk to occur.*
Almost Certain	*Expected to occur often*
Likely	*Will probably occur in time*
Moderate	*May occur occasionally.*
Unlikely	*Possible, but not expected*
Rare	*Unlikely to occur*
Consequence	
Insignificant	*Minor harm, no lost time.*
Minor	*Minor injury, slight damage.*
Significant	*Moderate injury/illness, significant damage*
Major	*Severe injury/illness, major damage, potential fatality*
Severe	*Death or catastrophic, widespread loss*
Draft by	Aaron Dooley
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Appendix 2: Bushfire Safety *(applies to all field activities in bushfire-prone areas)*

Pre-Travel Bushfire Readiness

Before leaving for any project site during bushfire season or in bushfire-prone landscapes, field staff must:

Mandatory Checks

- Review Site-Specific ERP, including fire triggers, evacuation routes, refuge points, and communication procedures.
- Complete all pre-travel WHS forms, vehicle checks, communications checks and any other pre-mobilisation checks.
- Confirm fuel levels are ~~full~~full, and vehicles are suitable for rapid evacuation if needed.

Fire & Weather Applications *(must check before departure)*

- **BOM Weather** – temperature, wind, thunderstorm risk.
- **Hazards Near Me** – fire alerts.
- **Lightning app-** lightning alerts
- **RFS: Fires Near Me** – fire behaviour and warnings.
- **Air Quality NSW** – smoke conditions.
- **Live Traffic NSW** – fire impacts on roads/closures.

Do not travel into:

- Emergency Warning or Watch & Act zones.
- Very Poor / Hazardous Air Quality zones.
- Areas under active fire threat or unpredictable plume behaviour.

On-Site Bushfire Safety Requirements

Mandatory Behaviours

- Conduct a fire-specific Toolbox Talk on arrival covering:
    - Current fire risk and conditions
    - Escape routes
    - Safe assembly point
    - Communication methods
    - Fire behaviour triggers
- Maintain continuous awareness of wind shifts, smoke, visibility, and temperature spikes.
- Maintain mandatory check-ins with the Communication Officer.
- Never work alone in a remote or isolated project location.

On-site Prohibitions

- **Do not** work in areas affected by active smoke columns, poor visibility, or unpredictable fire behaviour.
- **Do not** leave vehicles idling in long grass.
- **Do not** enter dense fuel load areas during heightened fire danger.

Required Equipment

- Charged mobile phone and In Reach *(remote sites).*
- First aid kit
- Fire extinguisher
- P2/N95 masks for smoke exposure.
- 10+ litres of water per person per day

Bushfire Cease-Work Triggers

Work must cease immediately if any of the following occur:

- Sudden increase in smoke, temperature, or wind direction change.
- Fire spotted within the vicinity *(for example, visual smoke, glow, ash fallout).*
- RFS “Watch & Act” or “Emergency Warning”, has been issued for the district.
- Visibility deteriorates to unsafe levels for travel or terrain navigation.
- Lightning detected within 30 seconds (30/30 rule).
- Air Quality reaches “Very Poor” or “Hazardous” categories.

Action When Trigger Occurs

1. Stop work, move to your vehicle and reassess safety.
2. Notify Communication Officer.
3. Determine if safe evacuation is required *(use ERP routes).*

Designated Safe Refuge Areas

Each project ERP must identify low-fuel refuge zones, which may include:

- Short grass clearings
- Bare earth or rocky areas
- Cleared firebreaks
- Roadsides without tree canopy cover

Field Lead **must** brief the team on:

- Refuge location

- Access routes *(primary & secondary)*
- Actions to shelter if evacuation becomes impossible

Evacuation Protocol

If evacuation is required:

- Follow ERP evacuation routes.
- Drive away from fire using the direction opposite to wind flow *(if known).*
- Avoid areas with heavy fuel loads, tree fall risk, or poor visibility.
- Confirm safe arrival with Communication Officer.

Cease-Work Authority

Any team member has the authority to cease work immediately if they detect fire-related risks or unsafe conditions.

Appendix 3: Lightning Safety Protocol *(applies to all field activities in storm-prone or exposed areas).*

Pre-Travel Lightning Readiness

Before departing for fieldwork in areas with potential thunderstorm activity, all field staff must complete the following:

Mandatory Checks

Review the Site-Specific ERP, including lightning triggers, evacuation routes, shelter points, and communication procedures.

- Complete all pre-travel WHS forms, vehicle checks, communication checks, and required pre-mobilisation tasks.
- Ensure vehicles are fuelled and ready for rapid evacuation, with reliable traction and safe parking options.
- Confirm all staff understand the 30/30 Lightning Rule and the site-specific cessation triggers.

Note:

STOP WORK immediately if the time between lightning flash and thunder is **30 seconds or less**.

Do NOT resume fieldwork until **30 minutes after the last thunder** is heard.

Storm & Weather Applications *(must check before departure)*

- **BOM Weather** – thunderstorm risk, severe weather warnings, wind, rainfall, and temperature.
- **Lightning Tracker App** – real-time lightning activity and storm approach.
- **Hazards Near Me** – severe storm alerts.
- **Live Traffic NSW** – road closures, fallen trees, storm-affected routes.
- **Emergency+ App** – location accuracy in emergencies.
- **Air Quality NSW** – smoke or storm-related particulates that reduce visibility.

Do not travel if:

- Thunderstorm warnings are active for the district.
- Lightning is detected approaching the work area on radar.
- Visibility is significantly reduced by rain, storm fronts, smoke or dust.
- Travel routes are affected by storm damage, fallen trees, or flooding indicators.

On-Site Lightning Safety Requirements

Mandatory Behaviours

- Conduct a lightning-specific Toolbox Talk on arrival covering:
  - Current storm conditions and forecast
  - Safe shelter locations
  - Primary and secondary evacuation routes
  - Communication procedures & check-in requirements
  - Lightning strike triggers (including the 30/30 rule)
- Maintain continuous weather monitoring, including cloud build-up, thunder sound, wind shifts, rainfall intensity and visibility.
- Conduct mandatory check-ins with the Communication Officer (as per ERP).
- Never work alone when thunderstorms are possible.
- Maintain constant communication capacity (UHF + mobile + InReach).

On-Site Prohibitions

When lightning risk exists:

- Do NOT continue fieldwork if thunder is heard – stop immediately.

- Do NOT shelter under trees, cliffs, or near tall isolated objects.
- Do NOT enter or work near waterbodies, creeks, or drainage lines.
- Do NOT handle metal tools or equipment (for example, GPS poles, star pickets, ladders, fencing wire, survey poles).
- Do NOT conduct aerial, elevated, or metal-assisted tasks, including:
  - EWP operation
  - Climbing
  - Drone operation
  - Use of tall tripods or antennae
    - Cease all work at height.

Required Equipment

- UHF radio + mobile phone + In Reach/PLB (remote locations).
- Fully charged battery packs.
- First aid kit.
- Weatherproof clothing (to remain warm during sheltering).
- Safe, enclosed vehicle available for immediate shelter

Lightning Cease-Work Triggers

Fieldwork must cease immediately if ANY of the following occur:

- Thunder is heard, regardless of distance.
- Lightning is visible, regardless of distance.
- Time between lightning and thunder ≤ 30 seconds (30/30 rule).
- Storm front moves toward the work zone on radar or visual observation.
- Winds increase suddenly, or heavy rain threatens visibility or safe footing.
- Any staff member feels unsafe or observes warning signs of storm escalation.

Actions When a Trigger Occurs

1. Stop work immediately – put down all tools and move away from conductive materials.
2. Proceed to vehicles or a safe enclosed structure (metal cabin of a vehicle is safest).
3. Notify Communication Officer that cease-work has been activated.
4. Reassess conditions from shelter (not in open areas).
5. Await "all clear":
    - Lightning must have ceased for at least 30 minutes
    - No thunder for 30 minutes
    - Storm is visibly moving away
6. Do **not** resume work until the Field Lead confirms conditions are safe.

Designated Safe Shelter Areas

The ERP must identify appropriate lightning-safe shelter locations such as:

- Fully enclosed vehicles (preferred)
- Solid buildings or sheds (if available)
- Depressions or low terrain - only if far from trees, metal and water

Avoid the following as shelter:

- Trees (single or cluster)
- Ridges, hilltops, exposed rock
- Scaffolding, metal structures
- Waterbodies or gullies

The Field Lead must brief the team on:

- Shelter points
- Evacuation routes
- Backup/safe alternative routes
- Assembly expectations

Evacuation Protocol *(Lightning)*

If evacuation is required due to storm severity:

- Follow ERP evacuation routes away from storm-intensified areas.
- Avoid roads with fallen tree risk, flooding or landslip indicators.
- Drive cautiously with headlights and hazards on during low visibility.
- Stop safely if visibility drops below safe travel thresholds.
- Confirm safe arrival with the Communication Officer immediately.

Cease-Work Authority

Any team member has the authority to stop work immediately if lightning is observed, thunder is heard, or if personal safety is compromised by storm activity.